

Vol 15 Chapter 2 — AC Graph as Research Instrument

Keeper (author pass — deep math/physics revision)

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Chapter 2 — AC Graph as Research Instrument

Level 1 — one sentence

The AC theorem graph (~2200 nodes, ~9800 edges, 55+ domains) is BST's research instrument: a queryable knowledge structure for finding shortest-path proofs across domain boundaries, with Casey's wall-routing principle (May 15 standing order) directing CIs to search the graph rather than push harder against direct obstructions.

Level 2 — graduate-physicist precision

2.1 Graph structure (recap)

Vol 14 Ch 7 covered the AC graph in detail. Methodologically: - Nodes: 2200+ theorems (T1-T2476+) - Edges: 9800+ directed derivations - Domains: 55+ - Tools: `play/toy_bst_explorer.py REPL`

2.2 Wall-routing principle

Casey's standing order (May 15): when you hit a wall, don't push harder — search the graph.

```
python3 play/toy_bst_explorer.py connect <blocked_concept> <target>
```

Returns alternative paths through other domains.

`/route` skill formalizes wall-routing as repeatable methodology.

2.3 Empirical: cross-domain bridges work

Documented examples across BST history: - Number theory $\leftrightarrow$ geometry (Heegner-Stark numbers via D_{-IV}^5) - Physics $\leftrightarrow$ math (Bergman kernel = Born rule) - Topology $\leftrightarrow$ algebra (K3 surface as Bridge Object) - $AC(0) \leftrightarrow$ curvature ($P \neq NP$ via Gauss-Bonnet)

When you have a wall in one domain, there is almost always a route through another.

2.4 Adding to the graph

Discipline: 1. Claim theorem ID via `/theorem claim` 2. Register node with domain, dependencies, proof status 3. Add edges to/from related theorems 4. Build verification toy (Vol 15 Ch 4)

Maintained by team (Lyra primary; Grace + Keeper + Elie support).

2.5 K-audit anchors

- Vol 14 Ch 7: AC graph structure
- `/route` skill (Casey standing tool)
- AC graph commitment_cycle_zone systematic backfill (Grace multi-day, completed)

Level 3 — 5th-grader accessibility

AC graph = the team's queryable knowledge structure. 2200+ theorems, 9800+ edges. Wall routing: when stuck on a problem, search the graph for a path through another domain. /route skill. Empirical: cross-domain bridges almost always exist. Discipline: claim theorem ID, register, build verification toy.

What comes next

Chapter 3 develops how to discover a new BST theorem.

Where to look this up

- BST: AC graph; /route skill; Vol 14 Ch 7